

A SIMPLE AUDIO AMPLIFIER With Dual Power Source

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This simple audio amplifier can be built at home with few components. You can use it at home or in a car with the 12V car battery as power source. The author's prototype is shown in Fig. 1 while the circuit diagram is shown in Fig. 2.

The circuit comprises step-down transformer X1, bridge rectifier BR1, 12V voltage regulator IC LM7812, amplifier IC LA4440, 20W loudspeaker

LS1, and few other components. Capacitors C1 and C7 connected across the supply terminals are used to minimise ripples/noise signals.

This audio amplifier with dual power source operates at 12V DC, which can be derived in two ways. The first uses a 230V AC primary to 15V, 1A step-down transformer X1. Its 15V AC secondary is connected to bridge rectifier BR1 for rectification of the voltage, which is filtered by ca-

pacitor C1. The rectified and filtered voltage is given to the regulator IC LM7812 to get regulated voltage of 12V for the circuit. Any ripples or noise in the voltage are filtered by capacitor C7.

The second, easier, way is to use a 12V, 7Ah battery. Switch S1 is used to select the supply either from the mains or from the battery, especially when the supply from mains is not available.

The LA4440 IC used is a two-channel audio amplifier with inbuilt dual channels. One of these is mono (bridge) mode and the second is dual (stereo) mode.

In dual mode, it produces 6 watts on each channel and in bridge mode it gives 19 watts output. The ripple rejection is 46dB. Normally, the IC works at 12V but it can handle up to 18V.

When using in dual channel mode, you get $6W + 6W = 12$ watts.

The circuit presented here is for mono/bridge mode operation to get 19W output. It has inbuilt over-voltage and surge-voltage protection.

While using mains, flip switch S1 towards mains side to disconnect the battery from circuit. When the battery is used, the components on the left side of dotted line are not required. In this case you just connect the 12V battery and flip switch S1 towards battery's side, which disconnects these components.

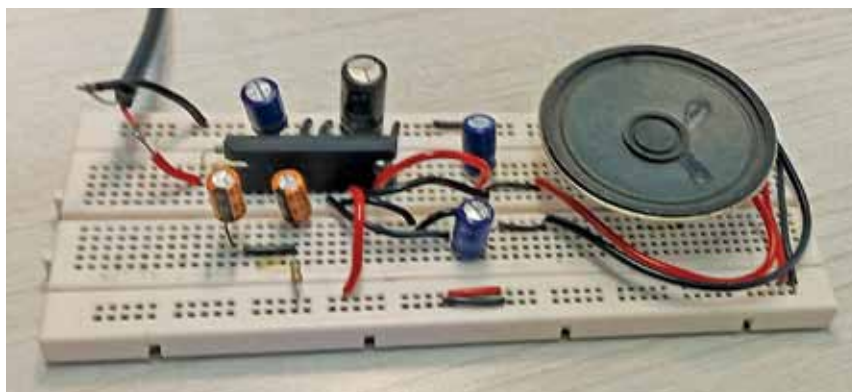


Fig. 1: Author's prototype on a breadboard

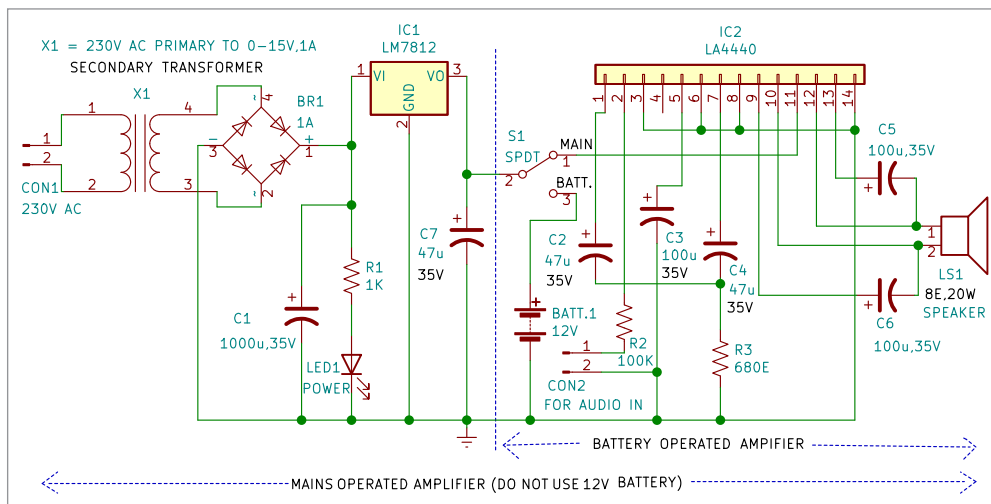


Fig. 2: Circuit diagram of the audio amplifier